2.7 Six Digits or Six Hundred

WORKSHEET

The question

STEP II 2023, Question 5

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Do you notice anything about the two lists? (Look at which fractions appear in both.)

STAGE 1 The first recipe stays at least 1

Covers: Part (i)

Ask yourself Rewrite x_{n+1} as $1 + \frac{1}{\text{something}}$. Then use induction.

STAGE 2 The error in the first recipe

Covers: Part (i)

Pause and try Write $x_{n+1}^2 - 2$ as a single fraction in terms of x_n. What do you notice about the numerator? How small must $\frac{1}{(x_n+1)^2}$ be?

STAGE 3 Ten steps

Covers: Part (i)

Ask yourself Apply the inequality ten times, starting from x_0. For the direction of the inequality, follow the sign of $x_n^2 - 2$: what happens to it at each step?

STAGE 4 The error in the second recipe

Covers: Part (ii)

Pause and try Write $y_{n+1} - \sqrt{2}$ as a single fraction. The numerator should remind you of a perfect square. Then use induction to show $y_n \geq 1$.

STAGE 5 Squaring the error

Covers: Part (ii)

Ask yourself Write $e_n = y_n - \sqrt{2}$. Using $y_n \geq 1$, bound e_{n+1} in terms of e_n^2.

Pause and try Prove $e_n \leq 2r^{2^n}$ for $n \geq 1$ by induction, where $r = \frac{\sqrt{2}-1}{2}$. Check the base case carefully.

STAGE 6 Six hundred digits

Covers: Part (ii)

Ask yourself How does r compare with $\frac{1}{4}$? Then compare 2^{10} with 10^3.

What's really going on

Newton's method

The second recipe is *Newton's method* applied to $f(y) = y^2 - 2$. Newton's method finds a root of f by repeatedly replacing a guess y by the point where the tangent at y crosses the axis:

$$y_{\text{new}} = y - \frac{f(y)}{f'(y)} = y - \frac{y^2 - 2}{2y} = \frac{y^2 + 2}{2y}.$$

Near a simple root α, a Taylor expansion shows that the new error is approximately $\frac{f''(\alpha)}{2f'(\alpha)} \times (\text{old error})^2$, which is why Newton's method converges quadratically. This particular case, "average y and $2/y$", was known to the Babylonians; a clay tablet from around 1700 BC, now called YBC 7289, gives $\sqrt{2}$ in base 60 with an error of less than one part in a million.

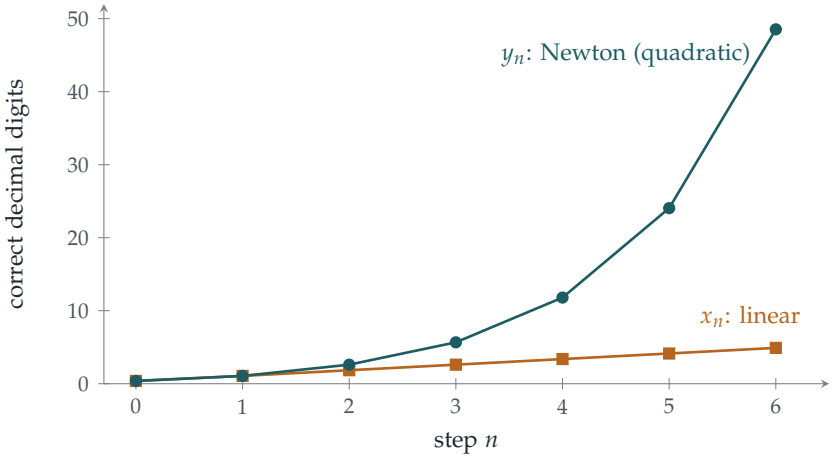


Figure 2.7.1 The number of correct decimal digits of $\sqrt{2}$ after n steps of each recipe. The linear recipe gains about 0.77 digits per step; Newton's method roughly doubles its number of correct digits at every step, reaching about 780 digits by $n = 10$.

Fixed points and their speed

Both recipes have the form $u_{n+1} = g(u_n)$ with $g(\sqrt{2}) = \sqrt{2}$, so $\sqrt{2}$ is a *fixed point*, as in the cobweb diagram of Section 2.2. Near a fixed point α, the error is multiplied by about $g'(\alpha)$ at each step. For the first recipe, $g(x) = \frac{x+2}{x+1}$ and $g'(\sqrt{2}) = -\frac{1}{(\sqrt{2}+1)^2} = -(3-2\sqrt{2}) \approx -0.17$: the minus sign explains the alternating signs and 0.17 explains the 0.77 digits per step ($\log_{10}(1/0.17) \approx 0.77$). For Newton's method $g'(\alpha) = 0$, so the linear term vanishes and the error is controlled by the next term, the square. Designing iterations with $g'(\alpha) = 0$ is the art of fast numerical methods.

Why the same fractions appear

The fractions $x_n = p_n/q_n$ satisfy $p_{n+1} = p_n + 2q_n$ and $q_{n+1} = p_n + q_n$, and $p_n^2 - 2q_n^2 = \pm 1$, alternating. These are the solutions of *Pell's equation* $p^2 - 2q^2 = \pm 1$, and they are the convergents of the continued fraction $\sqrt{2} = [1; 2, 2, 2, \ldots]$. They come from powers of $1 + \sqrt{2}$: $p_n + q_n\sqrt{2} = (1+\sqrt{2})^{n+1}$. Newton's step *squares* $p + q\sqrt{2}$ (try it: $(p+q\sqrt{2})^2 = p^2 + 2q^2 + 2pq\sqrt{2}$, and $\frac{p^2+2q^2}{2pq}$ is exactly Newton's formula applied to p/q). Squaring doubles the exponent, which is why $y_n = x_{2^n-1}$. The same pattern of "squaring" drives the iteration in Section 2.8.

Try this next

1 **Pell's equation.** Show that if $y_n = p/q$ in lowest terms with $p^2 - 2q^2 = 1$, then $y_{n+1} = \dfrac{p^2+2q^2}{2pq}$ also satisfies this equation. Check that $577^2 - 2 \times 408^2 = 1$.

2 **Cube roots.** Write down Newton's method for $f(y) = y^3 - 2$, and show that the new error is $\dfrac{(y_n - \alpha)^2(2y_n + \alpha)}{3y_n^2}$, where $\alpha = \sqrt[3]{2}$. Deduce that if $y_n > \alpha$ then the new error is at most $\frac{1}{\alpha}$ times the square of the old error. (*Hint:* use $\alpha^3 = 2$.)

3 **Division without dividing.** Show that Newton's method for $f(y) = \frac{1}{y} - a$ gives $y_{n+1} = y_n(2 - ay_n)$, which needs no division. If $e_n = 1 - ay_n$, show that $e_{n+1} = e_n^2$. (Computer processors use iterations like this to divide.)

4 **When Newton fails.** Apply Newton's method to $f(y) = y^{1/3}$, whose only root is 0. Show that each step doubles the distance from the root, whatever you start with. Which hypothesis of the error formula fails?

Further reading

- **Endre Süli & David Mayers, *An Introduction to Numerical Analysis*** (Cambridge University Press). The standard undergraduate text; Chapter 1 covers fixed-point iteration and Newton's method.
- **Nick Higham (ed.), *The Princeton Companion to Applied Mathematics*** (Princeton University Press). Short articles on Newton's method and numerical analysis, written by experts for a wide audience.
- **David Fowler & Eleanor Robson, "Square root approximations in Old Babylonian mathematics: YBC 7289 in context"**, *Historia Mathematica* 25 (1998).

The full guided solution is in Section 2.7:
https://beyondstep.keithmcnulty.org/ch02/newton-speed.html